

NimbusStack Internal Technical Overview

Confidential — Prepared by Ritika Vohra, Lead Platform Architect

1. Company Background

NimbusStack Technologies is a fictional cloud infrastructure company founded in 2019 by Devraj Malhotra and Priya Ashwin. The company is headquartered in the fictional city of Kestrel Bay. NimbusStack's flagship product is a container orchestration platform called FerroGrid, first released in March 2021.

2. FerroGrid Architecture

FerroGrid is composed of four core modules: the Quillion Scheduler, the Basalt Network Mesh, the Tundra Storage Layer, and the Ember Observability Suite. The Quillion Scheduler is responsible for placing workloads across nodes using a custom algorithm called Weighted Fracture Balancing (WFB), which was designed to reduce node hotspotting by 42 percent compared to FerroGrid's earlier round-robin scheduler.

The Basalt Network Mesh handles inter-pod communication and enforces network policies. It uses a proprietary protocol named Glyph Transport Protocol (GTP-7), which operates on port 6743 by default.

The Tundra Storage Layer provides persistent volume management and supports a maximum of 18 petabytes per cluster. It uses a technique called Snowline Replication, which maintains three synchronous replicas and two asynchronous replicas by default.

The Ember Observability Suite collects metrics, logs, and traces. It ships with a built-in anomaly detection model called SparkWatch, which flags anomalies using a rolling z-score threshold of 3.2.

3. Release History

FerroGrid version 1.0 was released in March 2021. Version 2.0, codenamed 'Halcyon', was released in November 2022 and introduced the Quillion Scheduler. Version 3.0, codenamed 'Marrow', was released in July 2024 and introduced GTP-7 as a replacement for the older Basalt Legacy Protocol. The most recent version, 3.5, codenamed 'Thistlewood', was released in January 2026 and added support for the Tundra Storage Layer's Snowline Replication feature.

4. Pricing Tiers

NimbusStack offers three pricing tiers for FerroGrid. The Sparrow tier costs 240 fictional credits per month and supports up to 10 nodes. The Kestrel tier costs 890 fictional credits per month and supports up to 75 nodes. The Condor tier is usage-based and has no node limit, billed at 4.5 fictional credits per node-hour.

5. Security Model

FerroGrid enforces a zero-trust security model called Vantablock. Vantablock requires every service-to-service call to present a short-lived token issued by the internal identity authority named Threshold Auth. Tokens issued by Threshold Auth expire after 90 seconds by default, and this value can be configured between 30 and 300 seconds. The security team, led by Nandini Rege, mandates that all clusters rotate their root signing key every 45 days.

6. Incident History

In September 2025, NimbusStack experienced an internal incident referred to as the 'Thornfield Outage', caused by a misconfigured Snowline Replication policy that led to a cascading failure across the Tundra Storage Layer in the fictional Kestrel Bay data center. The outage lasted 3 hours and 12 minutes and was resolved by rolling back to Snowline Replication configuration version 4.

7. Support and SLA

NimbusStack's support SLA guarantees a response time of 15 minutes for Condor tier customers, 2 hours for Kestrel tier customers, and 24 hours for Sparrow tier customers. The support team is led by Operations Manager Farrukh Siddiqui.